

References

- [1] “The LAPACK functions in GPU-accelerated linear algebra library (CULA),” <http://www.culatools.com/features/lapack/>.
- [2] “Apache design solutions,” white papers, www.apache.org.
- [3] *Assessing Product Reliability*. NIST/SEMATECH e-Handbook of Statistical Methods, ch. Chapter 8, <http://www.itl.nist.gov/div898/handbook/>.
- [4] “Cadence Design Systems Inc.” <http://www.cadence.com>.
- [5] “Cadence Virtuoso UltraSim Full-Chip Simulator,” http://www.cadence.com/products/cic/UltraSim_fullchip.
- [6] “Chandu Visweswariah,” http://domino.research.ibm.com/comm/research_people.nsf/pages/chandu.index.html.
- [7] “Computer Engineering International Training and Research Experience for Students (Ceitres@UCR),” <http://www.ee.ucr.edu/~stan/ceitres/Home.html>.
- [8] “Correlation clustering system,” <http://www.cs.brown.edu/~melsner/>.
- [9] “Critical Reliability Challenges for The International Technology Roadmap for Semiconductors (ITRS),” in International Sematech Technology Transfer Document 03024377A-TR, 2003.
- [10] “DAC panelists eye EDA in 2017,” <http://eetimes.eu/showArticle.jhtml?articleID=199901570>.
- [11] “David Atienza,” <http://people.epfl.ch/david.atienza>.
- [12] “Early Academic Outreach Program (EAOP),” <http://www.eaop.ucr.edu>.
- [13] “Edxact Inc.” http://www.edxact.com/cor_mis.html.
- [14] “EECS217: GPU Architecture and Parallel Programming,” http://www.ee.ucr.edu/~stan/courses/eecs217/eecs217_home.htm.
- [15] “Eetimes,” <http://www.eetimes.com>.

- [16] "Electronic Design Automation Lab at Tsinghua University, Beijing, China," <http://tiger.cs.tsinghua.edu.cn/>.
- [17] "Failure Mechanisms and Models for Semiconductor Devices," in JEDEC Publication JEP122-A, Jedec Solid State Technology Association, 2002.
- [18] "FLOTHERM," <http://www.flomerics.com/flotherm/>.
- [19] "Gary Smith EDA," <http://www.garysmitheda.com>.
- [20] "Graduate Division Academic Preparation & outreach programs," <http://www.graduate.ucr.edu/GROtoc.html>.
- [21] "IBM power grid benchmarks," <http://dropzone.tamu.edu/~pli/PGBench/>.
- [22] "International Workshop on Emerging Circuits and Systems (IWECS)," http://www.ee.ucr.edu/stan/iwecs_workshop/iwecs_workshop.html.
- [23] "International Workshop on Emerging Circuits and Systems (IWECS09)," http://www.ee.ucr.edu/stan/iwecs_workshop/iwecs_workshop.html.
- [24] "Mentoring Summer Research Internship Program (MSRIP)," <http://www.graduate.ucr.edu/MSRIP.html>.
- [25] "Mixed mode - mixed level circuit simulator - ngspice," <http://ngspice.sourceforge.net/>.
- [26] "Mixed-Signal Nanometer VLSI Research Lab," <http://www.mscad.ee.ucr.edu/>.
- [27] "National Science Foundation, Computer and Communication Foundation (CCF), Core Program," http://www.nsf.gov/funding/pgm_summ.jsp?pims_id=503220&org=CCF.
- [28] "NSF CCF core program," <http://www.nsf.gov/pubs/2011/nsf11557/nsf11557.htm>.
- [29] "SPICE - the general-purpose circuit program for analog integrated circuits," <http://bwrc.eecs.berkeley.edu/Courses/IcBook/SPICE/>.
- [30] "Superlu v3.0," <http://crd.lbl.gov/xiaoye/SuperLU/>.

- [31] “Synopsys Corporation,” <http://www.synopsys.com>.
- [32] “UiMOR – UC Riverside Model Order Reduction Tool Suite,” http://www.ee.ucr.edu/stan/project/uimor/uimor_main.htm.
- [33] “University of california’s new leadership excellence through advanced degrees (UC LEADS),” <http://www.ucop.edu/ucleads/riverside.htm>.
- [34] “International technology roadmap for semiconductors(itrs), 2003 edition,” 2001, <http://public.itrs.net>.
- [35] “International technology roadmap for semiconductors(itrs), 2004 update,” 2004, <http://public.itrs.net>.
- [36] “International technology roadmap for semiconductors (itrs) 2007 edition,” 2007, <http://public.itrs.net>.
- [37] “International technology roadmap for semiconductors (ITRS),” 2009 update, <http://public.itrs.net>.
- [38] Z. Bai, J. Demmel, J. Dongarra, A. Ruhe, and H. van der Vorst, *Templates for the Solution of Algebraic Eigenvalue Problems: A Practical Guide*. SIAM, Philadelphia, 2000.
- [39] N. Bell and M. Garland, “Cusp: Generic parallel algorithms for sparse matrix and graph computations,” 2010, version 0.2.0. [Online]. Available: <http://cusp-library.googlecode.com>
- [40] M. Benzi, C. D. Meyer, and M. Tuma, “A sparse approximate inverse preconditioner for the conjugate gradient method,” *SIAM J. Sci. Comput.*, vol. 17, pp. 1135–1149, September 1996.
- [41] G. Box, G. M. Jenkins, and G. C. Reinsel, *Time Series Analysis: Forecasting and Control*, 3rd ed. Prentice-Hall, 1994.
- [42] R. Bridson and W.-P. Tang, “Refining an approximate inverse,” *J. Comput. Appl. Math.*, vol. 123, pp. 293–306, November 2000.
- [43] J. Burns, “TSV-based 3D integration,” in *Three Dimensional System Integration*, A. Papanikolaou, D. Soudris, and R. Radojcic, Eds. Springer, November 2010, pp. 13–22.

- [44] R. H. Byrd, R. B. Schnabel, and G. A. Shultz, “A trust region algorithm for nonlinearly constrained optimization,” *SIAM Journal on Numerical Analysis*, vol. 24, no. 5, pp. pp. 1152–1170, 1987. [Online]. Available: <http://www.jstor.org/stable/2157645>
- [45] K. Carey, “A matter of degrees: Improving graduation rates in four-year colleges and universities,” in *The Education Trust*, May 2004.
- [46] Ceitres@UCR, “Computer Engineering International Training and Research Experience for Students (Ceitres@UCR),” <http://www.engr.ucr.edu/ceitres>.
- [47] F. Chen, Y. Cao, and W. Ren, “Distributed computation of the average of multiple time-varying reference signals,” in *Proceedings of the American Control Conference*, San Francisco, CA, July 2011, pp. 1650–1655.
- [48] —, “Distributed average tracking of multiple time-varying reference signals with bounded derivatives,” *IEEE Transactions on Automatic Control*, vol. PP, p. 1, 2012.
- [49] E. K. P. Chong and S. H. Zak, *An Introduction to Optimization*, 2nd ed. New York: Wiley-Interscience, 2001.
- [50] E. Chow, “Parallel implementation and practical use of sparse approximate inverse preconditioners with a priori sparsity patterns,” *Int. J. High Perform. Comput. Appl.*, vol. 15, pp. 56–74, February 2001.
- [51] T. A. Davis, *Direct Methods for Sparse Linear Systems*. SIAM, Philadelphia, 2006.
- [52] T. A. Davis and Y. Hu, “The University of Florida Sparse Matrix Collection,” *ACM Transactions on Mathematical Software*, 2011.
- [53] J. W. Demmel, J. R. Gilbert, and X. S. Li, “An asynchronous parallel supernodal algorithm for sparse gaussian elimination,” *SIAM J. Matrix Analysis and Applications*, vol. 20, no. 4, pp. 915–952, 1999.
- [54] Q. Du, V. Faber, and M. Gunzburger, “Centroidal voronoi tessellations: Applications and algorithms,” *SIAM Review*, vol. 41, no. 4, pp. 637–676, 1999.
- [55] M. B. Eljamal, S. W. Pang, and S. J. Edington, “Gaining international competence: a multi-faceted approach to international engineering education,” in *Proc. 2005 American Society for Engineering Education Annual Conference & Exposition*, 2005.

- [56] C. A. Floudas, *Nonlinear and Mixed-Integer Optimization: Fundamentals and Applications (Topics in Chemical Engineering)*. Oxford University Press, 1995.
- [57] J. A. Frechtling, “User-friendly handbook for project evaluation: Science, mathematics, engineering and technology education,” in *National Science Foundation Publication 93-152*, 1993.
- [58] M. Frigo and S. G. Johnson, “FFTW,” MIT, Tech. Rep., 2009, <http://www.fftw.org/>.
- [59] O. Gay, D. Coeurjolly, and N. Hurst, “Libaffa: C++ affine arithmetic library for gnu/linux,” May 2005, <http://savannah.nongnu.org/projects/libaa/>.
- [60] P. E. Gill, W. Murray, Michael, and M. A. Saunders, “Snopt: An sqp algorithm for large-scale constrained optimization,” *SIAM Journal on Optimization*, vol. 12, pp. 979–1006, 1997.
- [61] S. Hebel, “A public research university succeeds because its low-income students do,” in *Chronicle of Higher Education*, March 2007.
- [62] J. R. Humphrey, D. K. Price, K. E. Spagnoli, A. L. Paolini, and E. J. Kelmelis, “CULA: hybrid GPU accelerated linear algebra routines,” in *SPIE Defense and Security Symposium (DSS)*, April 2010.
- [63] “Intel core 2 extreme microprocessor QX 9000 series,” Intel Co., 2009, <http://download.intel.com/design/processor/datashts/318726.pdf>.
- [64] “Intel pentium processor e5200 series specifications,” Intel Co., <http://ark.intel.com/Product.aspx?id=37212>.
- [65] “International technology roadmap for semiconductors(itrs) 2005, 2006 update,” 2006, <http://public.itrs.net>.
- [66] “International technology roadmap for semiconductors (ITRS) 2008 edition,” 2008, <http://public.itrs.net>.
- [67] “International technology roadmap for semiconductors (ITRS), 2010 update,” 2010, <http://public.itrs.net>.
- [68] “International technology roadmap for semiconductors (ITRS), 2010 update,” 2010, <http://public.itrs.net>.

- [69] “International technology roadmap for semiconductors (ITRS), 2011 edition,” 2011, <http://public.itrs.net>.
- [70] L. Ju, Q. Du, and M. Gunzburger, “Probabilistic methods for centroidal voronoi tessellations and their parallel implementations,” *Parallel Computing*, vol. 28, pp. 1477–1500, October 2002.
- [71] C. Ke, M. J. Ablowitz, S. H. Davis, E. J. Hinch, A. Iserles, J. Ockendon, and P. J. Olver, *Matrix Preconditioning Techniques and Applications; electronic version*. Cambridge: Cambridge Univ. Press, 2005.
- [72] C. Loretz, “Looking beyond the borders: A project director’s handbook of best practices for international research experiences for undergraduates,” in *NSF Workshop on Best Practices for Managing International REU Site Programs*, rev. 3/22/02, 2001.
- [73] M. Lynch and J. Engle, “Big Gaps, Small Gaps: Some colleges and universities do better than others in graduating african american students,” The Education Trust, Washington, DC, Tech. Rep., August 2010, <http://www.edtrust.org/dc/publication/big-gaps-small-gaps-in-serving-hispanic-students>.
- [74] —, “Big Gaps, Small Gaps: Some colleges and universities do better than others in graduating hispanic students,” The Education Trust, Washington, DC, Tech. Rep., August 2010, <http://www.edtrust.org/dc/publication/big-gaps-small-gaps-in-serving-african-american-students..>
- [75] “MCNC benchmark circuit placements,” <http://vlsicad.ucsd.edu/GSRC/bookshelf/Slots/Placement/>.
- [76] R. Nath, S. Tomov, and J. Dongarra, “An improved MAGMA GEMM for Fermi GPUs,” 2010.
- [77] “Science and engineering indicators,” 2006, two volumes. Arlington, VA: National Science Foundation (volume 1, NSB 06-01; volume 2, NSB 06-01A).
- [78] “Nangate open cell library,” <http://www.nangate.com/>.
- [79] R. Olfati-Saber and R. M. Murray, “Consensus problems in networks of agents with switching topology and time-delays,” *IEEE Transactions on Automatic Control*, vol. 49, pp. 1520–1533, 2004.

- [80] R. Patti, “3D integration: New opportunities for advanced packaging,” in *Electrical Performance of Electronic Package and Systems (EPEPS)*, October 2011, pp. 1–41.
- [81] “Predictive Technology Model,” <http://www.eas.asu.edu/~ptm/>.
- [82] Y. Saad, “Preconditioning techniques for nonsymmetric and indefinite linear systems,” *J. Comput. Appl. Math.*, vol. 24, pp. 89–105, 1988.
- [83] B. Smith and P. E. Bjorstad, *Domain Decomposition – Parallel Multilevel Methods for Elliptic Partial Differential Equations*. Cambridge University Press, <http://www.itl.nist.gov/div898/handbook/>.
- [84] “CCS timing technical white paper,” Synopsys Inc., http://www.opensourceliberty.org/ccspaper/ccs_timing_wp.pdf.
- [85] “UMFPACK,” <http://www.cise.ufl.edu/research/sparse/umfpack/>.
- [86] W. Wolf, *Modern VLSI Design: System-on-Chip Design*, 3rd ed. Prentice Hall, 2002.
- [87] J. Zhu and S. Calman, “Symbolic pointer analysis revisited,” in *Proceedings of the ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)*, June 2004.